Highlights

A Beta-evidential framework models epistemic uncertainty in multi-label ECG.

APEB resolves non-random interval missingness via clinical presence gating.

One-way logic loss cuts clinical contradictions from 0.88% to 0.68%.

Class-balanced probing rescues cross-center transfer without mV bias.

HE-DVR injects hard symbolic anchors into a 3B MLLM for grounded QA.